

NAME: PAVNI GULYANI

BATCH - 16 BTECH CSE

HDOC - 31

1. Algorithm

1. Start.
 2. Declare a 2D array *a*, a row-sum array *rowSum*, and variables for rows, columns, choice and sum.
 3. Input the number of rows and columns.
 4. Input the elements of the matrix.
 5. Display the menu.
 6. Read the user's choice.
 7. If choice is **1**, calculate and display the sum of all matrix elements.
 8. If choice is **2**, calculate the sum of each row, store the results in *rowSum[]*, and display them.
 9. If choice is **3**, calculate and display the transpose of the matrix.
 10. If choice is **4**, exit the program.
 11. If an invalid choice is entered, display **Invalid choice**.
 12. Repeat the menu until choice **4** is selected.
 13. Stop.
-

2. C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], rowSum[10];
```

```
int rows, cols;
```

```
int i, j, choice, sum;
```

```
printf("Enter the number of rows and columns: ");
```

```
scanf("%d %d", &rows, &cols);
```

```
printf("Enter the matrix elements:\n");
```

```
for (i = 0; i < rows; i++)
```

```
{
```

```
    for (j = 0; j < cols; j++)
```

```
    {
```

```
        scanf("%d", &a[i][j]);
```

```
    }
```

```
}
```

```
do
```

```
{
```

```
    printf("\nMenu:\n");
```

```
    printf("1. Find sum of all elements\n");
```

```
    printf("2. Find sum of each row\n");
```

```
    printf("3. Find transpose of matrix\n");
```

```
    printf("4. Exit\n");
```

```
    printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
case 1:
```

```
    sum = 0;
```

```
    for (i = 0; i < rows; i++)
```

```
    {
```

```
        for (j = 0; j < cols; j++)
```

```
        {
```

```
            sum = sum + a[i][j];
```

```
        }
```

```
    }
```

```
    printf("Sum of all elements = %d\n", sum);
```

```
    break;
```

```
case 2:
```

```
    printf("Row sums:\n");
```

```
    for (i = 0; i < rows; i++)
```

```
    {
```

```
        rowSum[i] = 0;
```

```
    for (j = 0; j < cols; j++)
    {
        rowSum[i] = rowSum[i] + a[i][j];
    }

    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
}
```

```
printf("The row-sum array contains: [");
```

```
for (i = 0; i < rows; i++)
{
    printf("%d", rowSum[i]);

    if (i < rows - 1)
        printf(", ");
}
```

```
printf("]\n");
break;
```

case 3:

```
printf("Transpose of the matrix:\n");
```

```
for (j = 0; j < cols; j++)
```

```

    {
        for (i = 0; i < rows; i++)
        {
            printf("%d ", a[i][j]);

        }

        printf("\n");
    }

    break;

case 4:
    printf("Exiting the program...\n");
    break;

default:
    printf("Invalid choice!\n");
}

} while (choice != 4);

return 0;
}

```

3. Compilation

Save the program as:

hdoc_day31.c

Open the terminal and compile using:

```
gcc hdoc_day31.c -o hdoc_day31
```

If there are **no errors**, the program is successfully compiled.

4. Execution

Run the program using:

```
./hdoc_day31
```

5. Execution / Sample Output

For the given matrix:

Enter the number of rows and columns: 2 3

Enter the matrix elements:

1 2 3

4 5 6

Menu:

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 1

Sum of all elements = 21

Menu:

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 2

Row sums:

Sum of row 1 = 6

Sum of row 2 = 15

The row-sum array contains: [6, 15]

Menu:

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 3

Transpose of the matrix:

1 4

2 5

3 6

Menu:

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 4

Exiting the program...

6. Test Case

Input	Expected Output
Rows = 2, Columns = 3 Matrix accepted	
1 2 3 4 5 6	Matrix stored
Choice = 1	Sum = 21
Choice = 2	Row sums = [6, 15]
Choice = 3	Transpose = 1 4 / 2 5 / 3 6
Choice = 4	Program exits

```
(kali@kali)-[~/Desktop]
$ gcc sample40.c -o sample40

(kali@kali)-[~/Desktop]
$ ./sample40
Enter the number of rows and columns: 3 4
Enter the matrix elements:
1 3 4 5 6 7 8
2 5 6 7 8 9 10
3 6 7 8 9 10 11
4
Menu:
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 67

Menu:
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 10
Sum of row 2 = 26
Sum of row 3 = 31
The row-sum array contains: [10, 26, 31]

Menu:
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
5 2
6 2
7 23
8 4

Menu:
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 4
Exiting the program...
```